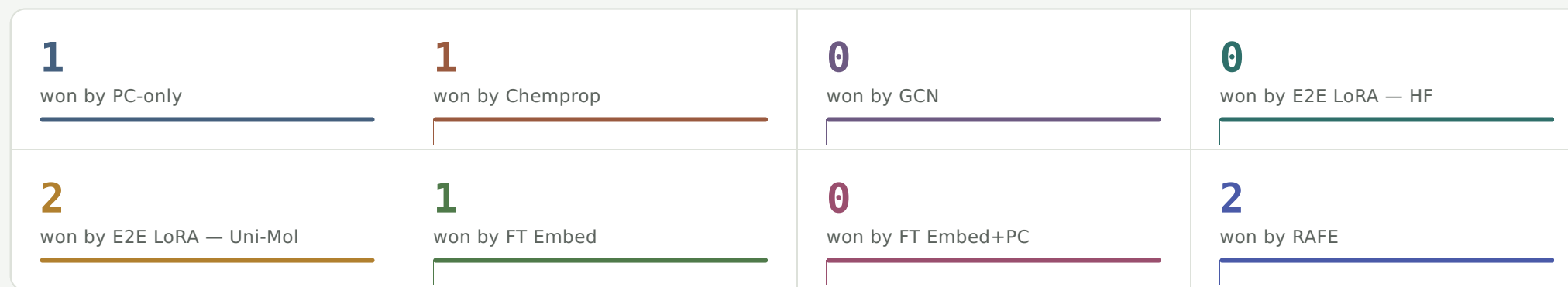
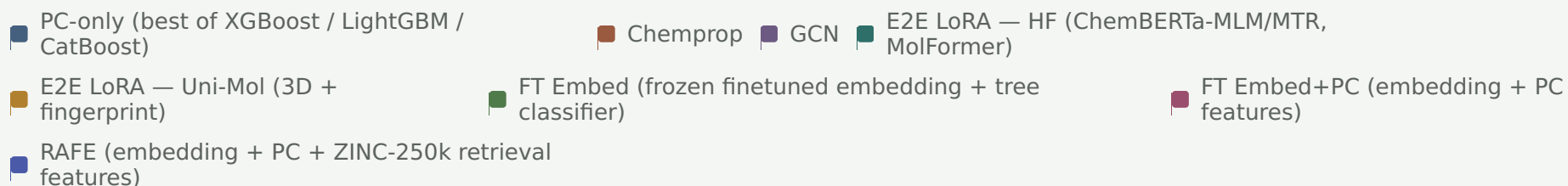


Every method, every dataset, every metric run so far

PC-only tree ensembles, Chemprop, GCN, end-to-end LoRA finetuning (HF CLM path and Uni-Mol path), FT Embed(+PC), and RAFE (retrieval-augmented features from ZINC-250k) compared head-to-head across all 7 benchmark datasets on the metrics each task actually reports. Best value in each dataset is highlighted. All 7 datasets now have a complete run across every method, including RAFE.

**Classification datasets**

MCC · AUC · Accuracy · Precision · Recall · F1-macro · F1-micro

BACE

BINARY · B-SECRETASE 1 (BACE-1) INHIBITION

METHOD	MCC	AUC	ACCURACY	PRECISION	RECALL	F1-MACRO	F1-MICRO
 PC-only CatBoost	0.498	0.850	0.724	0.746	0.751	0.723	0.724
 Chemprop	0.528	0.855	0.763	0.758	0.770	0.759	0.763
 GCN	0.447	0.795	0.717	0.719	0.729	0.714	0.717
 E2E LoRA — HF MolFormer, Focal Loss	0.566	0.868	0.783	0.777	0.789	0.779	0.783
 E2E LoRA — Uni-Mol Weighted Loss best	0.623	0.891	0.822	0.823	0.801	0.808	0.822
 FT Embed MolFormer (FL), CatBoost	0.557	0.869	0.763	0.773	0.784	0.762	0.763
 FT Embed+PC MolFormer (WL), LightGBM	0.550	0.857	0.763	0.769	0.781	0.762	0.763
 RAFE MolFormer (WL), LightGBM	0.561	0.858	0.770	0.774	0.787	0.768	0.770

BBBP

BINARY · BLOOD-BRAIN BARRIER PENETRATION

METHOD	MCC	AUC	ACCURACY	PRECISION	RECALL	F1-MACRO	F1-MICRO
 PC-only LightGBM	0.759	0.937	0.885	0.916	0.846	0.867	0.885
 Chemprop	0.620	0.918	0.807	0.799	0.821	0.801	0.807
 GCN	0.694	0.924	0.859	0.874	0.822	0.839	0.859
 E2E LoRA — HF MolFormer, Focal Loss	0.767	0.929	0.891	0.913	0.856	0.874	0.891
 E2E LoRA — Uni-Mol Weighted Loss	0.774	0.943	0.896	0.890	0.885	0.887	0.896
 FT Embed MolFormer (FL), XGBoost	0.787	0.935	0.901	0.914	0.873	0.888	0.901
 FT Embed+PC MolFormer (FL), LightGBM	0.789	0.934	0.901	0.920	0.870	0.887	0.901
 RAFE MolFormer (FL), XGBoost best	0.798	0.940	0.906	0.918	0.881	0.895	0.906

FLAVOR					5-CLASS · CANONICALIZED TASTE			
METHOD	MCC	AUC	ACCURACY	PRECISION	RECALL	F1-MACRO	F1-MICRO	
PC-only XGBoost	0.808	0.972	0.890	0.733	0.798	0.761	0.890	
Chemprop best	0.809	0.974	0.899	0.796	0.835	0.809	0.899	
GCN	0.798	0.973	0.893	0.834	0.800	0.814	0.893	
E2E LoRA — HF MolFormer, Weighted Loss	0.802	0.975	0.893	0.867	0.749	0.774	0.893	
E2E LoRA — Uni-Mol Weighted Loss	0.795	0.973	0.885	0.747	0.848	0.783	0.885	
FT Embed ChemBERTa-MTR (FL), XGBoost	0.793	0.953	0.881	0.724	0.759	0.738	0.881	
FT Embed+PC MolFormer (WL), CatBoost	0.801	0.958	0.888	0.756	0.793	0.772	0.888	
RAFE MolFormer (WL), CatBoost	0.804	0.963	0.893	0.868	0.749	0.774	0.893	

DILI					BINARY · DRUG-INDUCED LIVER INJURY			
METHOD	MCC	AUC	ACCURACY	PRECISION	RECALL	F1-MACRO	F1-MICRO	
PC-only CatBoost	0.765	0.914	0.875	0.894	0.870	0.872	0.875	
Chemprop	0.588	0.893	0.792	0.800	0.788	0.788	0.792	
GCN	0.728	0.889	0.854	0.880	0.849	0.850	0.854	
E2E LoRA — HF ChemBERTa-MTR, Weighted Loss	0.770	0.911	0.885	0.886	0.885	0.885	0.885	
E2E LoRA — Uni-Mol Focal Loss	0.708	0.926	0.854	0.855	0.853	0.854	0.854	
FT Embed ChemBERTa-MLM (WL), CatBoost	0.673	0.893	0.833	0.844	0.830	0.831	0.833	
FT Embed+PC Uni-Mol (FL), LightGBM	0.703	0.873	0.833	0.879	0.826	0.826	0.833	
RAFE best Uni-Mol (FL), XGBoost	0.790	0.875	0.885	0.910	0.880	0.882	0.885	

hERG

BINARY · CARDIAC ION-CHANNEL INHIBITION

METHOD	MCC	AUC	ACCURACY	PRECISION	RECALL	F1-MACRO	F1-MICRO
 PC-only XGBoost	0.604	0.874	0.802	0.802	0.802	0.802	0.802
 Chemprop	0.558	0.854	0.779	0.779	0.779	0.779	0.779
 GCN	0.510	0.833	0.755	0.755	0.755	0.755	0.755
 E2E LoRA — HF MolFormer, Weighted Loss	0.573	0.869	0.787	0.787	0.787	0.787	0.787
 E2E LoRA — Uni-Mol Weighted Loss best	0.609	0.881	0.804	0.805	0.804	0.804	0.804
 FT Embed Uni-Mol (FL), LightGBM	0.580	0.864	0.790	0.790	0.790	0.790	0.790
 FT Embed+PC Uni-Mol (FL), LightGBM	0.602	0.871	0.801	0.801	0.801	0.801	0.801
 RAFE Uni-Mol (FL), LightGBM	0.589	0.871	0.794	0.794	0.794	0.794	0.794

Regression datasets

R² · MAE · Spearman

CACO2

REGRESSION · CACO-2 PERMEABILITY (LOG PAPP)

METHOD	R2	MAE	SPEARMAN
 PC-only best CatBoost	0.744	0.273	0.853
 Chemprop	0.534	0.359	0.802
 GCN	0.244	0.491	0.733
 E2E LoRA — HF ChemBERTa-MTR, Huber	0.622	0.340	0.789
 E2E LoRA — Uni-Mol Huber	0.411	0.432	0.580
 FT Embed ChemBERTa-MTR, CatBoost	0.667	0.318	0.821
 FT Embed+PC ChemBERTa-MTR, LightGBM	0.671	0.308	0.824
 RAFE ChemBERTa-MTR (Huber), CatBoost	0.663	0.324	0.835

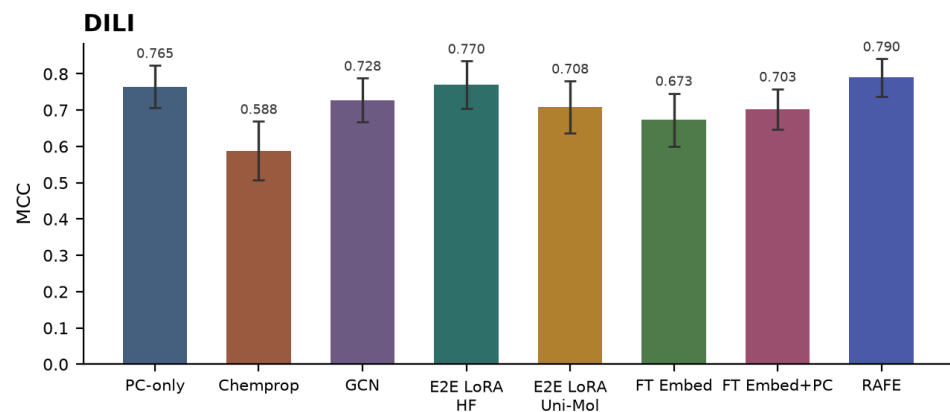
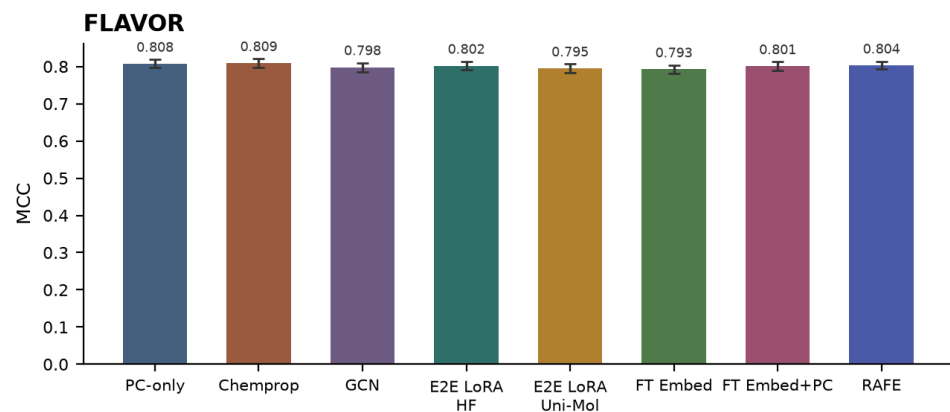
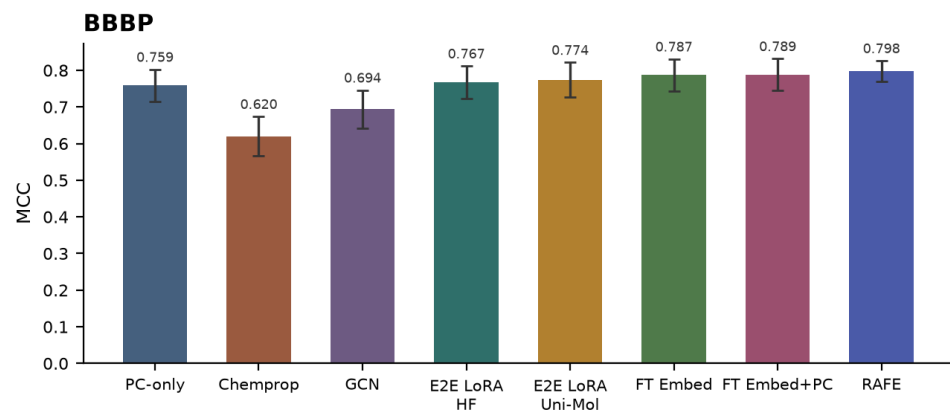
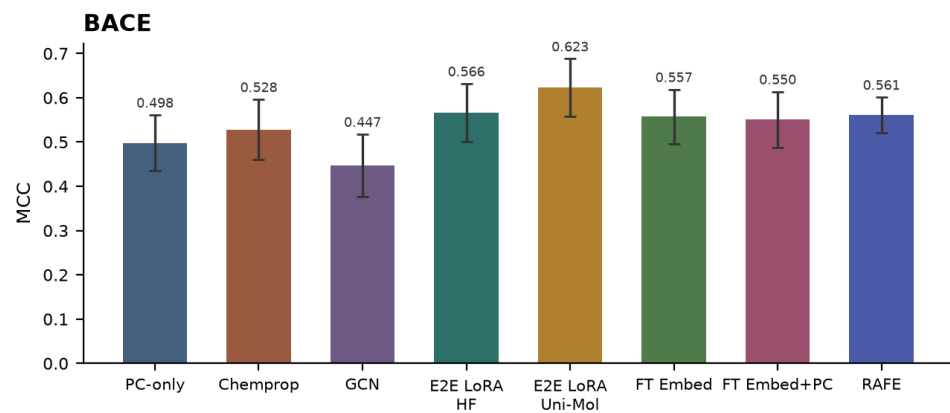
HALF_LIFE

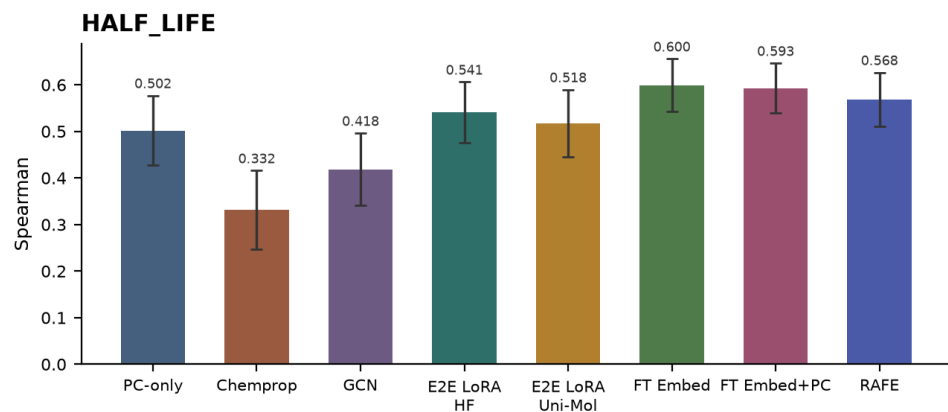
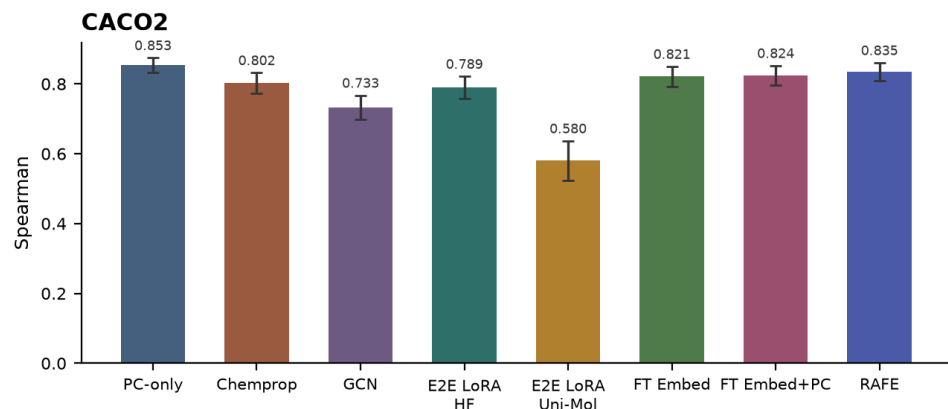
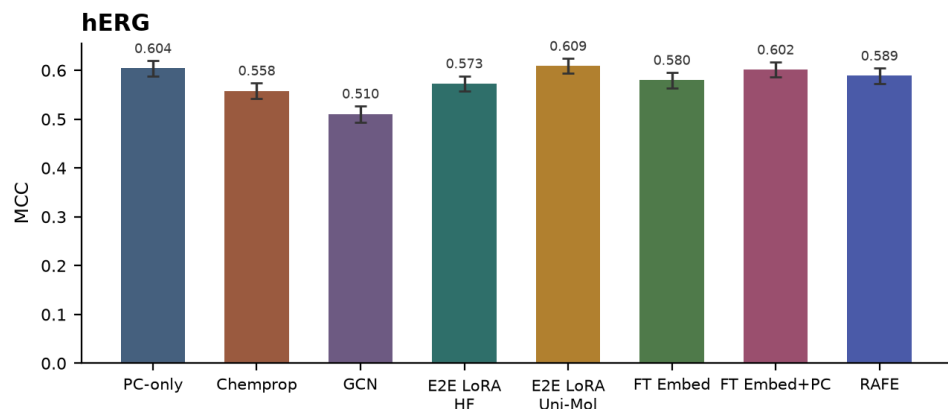
REGRESSION · DRUG HALF-LIFE (HOURS, LOG1P TARGET)

METHOD	R2	MAE	SPEARMAN
 PC-only XGBoost	0.110	5.686	0.502
 Chemprop	-0.020	6.325	0.332
 GCN	0.114	5.928	0.418
 E2E LoRA — HF ChemBERTa-MTR, Huber	0.072	6.367	0.541
 E2E LoRA — Uni-Mol Huber	-0.072	6.937	0.518
 FT Embed best MolFormer, XGBoost	0.083	5.639	0.600
 FT Embed+PC MolFormer, XGBoost	0.090	5.593	0.593
 RAFE MolFormer (Huber), XGBoost	0.090	5.557	0.568

Mean $\pm$ standard error, by dataset

primary metric (MCC / Spearman) · bar = point estimate · whisker = bootstrap SE





DATASET	METRIC	PC-ONLY	CHEMPROP	GCN	E2E LORA – HF	E2E LORA – UNI-MOL	FT EMBED	FT EMBED+PC	RAFE
BACE	MCC	0.498 ± 0.063	0.528 ± 0.068	0.447 ± 0.071	0.566 ± 0.066	0.623 ± 0.065	0.557 ± 0.062	0.550 ± 0.063	0.561 ± 0.040
BBBP	MCC	0.759 ± 0.043	0.620 ± 0.054	0.694 ± 0.052	0.767 ± 0.045	0.774 ± 0.048	0.787 ± 0.044	0.789 ± 0.043	0.798 ± 0.028
FLAVOR	MCC	0.808 ± 0.011	0.809 ± 0.011	0.798 ± 0.012	0.802 ± 0.011	0.795 ± 0.011	0.793 ± 0.011	0.801 ± 0.011	0.804 ± 0.010
DILI	MCC	0.765 ± 0.058	0.588 ± 0.081	0.728 ± 0.060	0.770 ± 0.065	0.708 ± 0.072	0.673 ± 0.073	0.703 ± 0.055	0.790 ± 0.051
hERG	MCC	0.604 ± 0.016	0.558 ± 0.016	0.510 ± 0.017	0.573 ± 0.015	0.609 ± 0.015	0.580 ± 0.016	0.602 ± 0.015	0.589 ± 0.015
CACO2	Spearman	0.853 ± 0.021	0.802 ± 0.029	0.733 ± 0.034	0.789 ± 0.032	0.580 ± 0.057	0.821 ± 0.028	0.824 ± 0.028	0.835 ± 0.026
HALF_LIFE	Spearman	0.502 ± 0.075	0.332 ± 0.084	0.418 ± 0.078	0.541 ± 0.065	0.518 ± 0.072	0.600 ± 0.056	0.593 ± 0.053	0.568 ± 0.058

Every value carries its within-dataset bootstrap standard error (10,000 resamples on that dataset's own test set, normal-approximation). PC-only/Chemprop/GCN reuse the CIs computed during those runs (CACO2's PC-only was backfilled separately after its Optuna-budget rerun); E2E LoRA (HF and Uni-Mol) and FT Embed/FT Embed+PC whiskers were all computed by reloading each best saved checkpoint/model (including BACE/BBBP/FLAVOR's pre-existing `EffiChem_Extras` models, which predate this session and had no CI until now) and bootstrapping its test-set predictions the same way. RAFE whiskers for `herg/dili/caco2/half_life` are the same full reconstruct-and-rebootstrap treatment; RAFE whiskers for `bace/bbbp/flavor` come from a pre-existing approximate-CI file instead (see reading notes above) since those runs' models/features were not retained. Regression "best" is judged by Spearman, not R2, throughout. Where two values' $\pm$ SE bands overlap, the point-estimate ranking between them is not statistically confirmed — per the current convention, treat overlap within ± 1 SE (not the wider ± 1.96 SE/95% band mentioned here) as the bar for "not yet a confirmed difference."

Reading notes

- All methods now have bootstrapped 95% CIs (10,000 resamples, normal-approximation) on every dataset — see the "Mean $\pm$ standard error, by dataset" section below for the whisker view. A close point-estimate "win" (e.g. a LoRA method edging past PC-only) should still be read against the whisker overlap there before treating it as statistically confirmed.
- E2E LoRA columns show the best-performing loss variant (Focal / Weighted / Huber) and base model for that dataset; the sub-label under each method name names which one.
- **FT Embed** and **FT Embed+PC** freeze the finetuned CLM as an embedding extractor and train a fresh XGBoost/LightGBM/CatBoost search on top — either the embedding alone, or embedding + the same 473-dim PC feature vector used by PC-only. Best of HF (6 sub-models $\times$ 2 losses) and Uni-Mol is shown (by Spearman for regression, MCC for classification), sub-labelled with the winner, and every entry now has a bootstrapped CI (including BACE/BBBP/FLAVOR's pre-existing `EffiChem_Extras` results, computed by reloading the saved model and its exact test feature matrix). Uni-Mol never got an FT+PC variant for the original 4 datasets (HF only). FT Embed+PC trails PC-only almost everywhere — concatenating a 384–2560-dim frozen, mean-pooled embedding onto 473 PC features gives tree ensembles a much higher-dimensional, less sample-efficient search than PC-only's curated feature set, especially on the smaller datasets.
- **RAFE** (Retrieval-Augmented Feature Extraction) concatenates [CLM embedding + PC features + 57 ZINC-250k retrieval features per model] — nearest-neighbor physicochemical stats (logP/QED/SAS, $k=10/20$), neighborhood density, a PCA-32 centroid, and Tanimoto structural similarity, retrieved via a GPU FAISS cosine index over all 249,455 ZINC-250k molecules. Best of HF and Uni-Mol is shown per dataset (by MCC/Spearman), same as FT Embed. RAFE now beats every other method on **BBBP** (MCC 0.798) and **DILI** (MCC 0.790, Uni-Mol) — the only two datasets where the extra retrieval context clearly helps; elsewhere it lands close to FT Embed+PC, consistent with retrieval features adding little once the model already has a strong embedding plus curated PC descriptors. BACE/BBBP/FLAVOR's CI comes from a pre-existing `EffiChem_Extras` aggregate (`bootstrap_ci_approx.json`) rather than this session's reconstruct-and-rebootstrap method, because those RAFE runs' saved models and ZINC

retrieval-feature CSVs were not retained on disk — the approx CI is normal-approximation SE from the same resampling family but was not independently re-verified this session, so treat old-dataset RAFE whiskers as slightly less certain than the rest. herg/dili/caco2/half_life's RAFE CI is the full 10,000-resample bootstrap on freshly reloaded models, same as everywhere else.

- MolFormer's bundled remote code uses Performer-style random-feature attention; evaluation now forces `deterministic_eval=true` so results are reproducible run-to-run, matching the originally trained checkpoints' saved projection weights.
- **Verification audit (2026-07-24):** every mean/SE for herg/dili/caco2/half_life was re-checked against its ground-truth CI file and all match (one immaterial exception noted below); re-run at ± 1 SE, none of the 4 new datasets show a CLM-pipeline method clearing PC-only (every point-estimate winner overlaps PC-only's band) — the authoritative answer for Phase 9's reporting filter, though the dashboard's "best" badges above remain point-estimate only.